

Personal Information

Family Name: Tortorelli

First Name: Luca

Affiliation: Universitaet Sternwarte der Ludwigs-Maximilians-Universitaet Muenchen, Scheinerstr. 1, 81679, Munich, Germany

Email Address: Luca.Tortorelli@physik.lmu.de

Github: <https://github.com/torluca>

Education and Career

University Education

Ph.D.: Institute for Particle Physics and Astrophysics, ETH Zurich, Switzerland 09/2016 - 07/2021

Master's Degree: Dipartimento di Fisica "Ettore Pancini", Università degli Studi di Napoli "Federico II", Italy 10/2013 - 07/2016, Summa cum Laude

Bachelor's Degree: Dipartimento di Fisica "Ettore Pancini", Università degli Studi di Napoli "Federico II", Italy 10/2009 - 06/2013, Summa cum Laude

Title of Ph.D. Thesis: "Measurement of galaxy population properties with forward-modeling and Approximate Bayesian computation"

Names and Affiliations of Ph.D. Thesis Supervisors: Prof. Dr. Alexandre Refregier, Institute for Particle Physics and Astrophysics, ETH Zürich, Wolfgang-Pauli-Str. 27, 8093 Zürich, Switzerland

Prof. Dr. Adam Amara, Institute of Cosmology & Gravitation, University of Portsmouth, Dennis Sciamia Building, Burnaby Road, Portsmouth PO1 3FX, UK

Title of Master's Thesis: "The Kormendy relation of galaxies in clusters of the CLASH-VLT survey"

Names and Affiliations of Master's Thesis Supervisors: Dr. Amata Mercurio, Osservatorio Astronomico di Capodimonte, Via Moiarello 16, 80131, Naples, Italy

Prof. Maurizio Paolillo, Dipartimento di Fisica "Ettore Pancini", Università degli Studi di Napoli "Federico II", via Cintia, 80126, Naples, Italy

Title of Bachelor's Thesis: "Ellipticity Measurements in KiDS Survey Images Aimed at The Study of Gravitational Lensing"

Names and Affiliations of Master's Thesis Supervisors: Prof. Dr. Massimo Capaccioli, Osservatorio Astronomico di Capodimonte, Via Moiarello 16, 80131, Naples, Italy

Prof. Nicola Napolitano, School of Physics and Astronomy, Sun Yat-sen University Zhuhai Campus, 2 Daxue Road, Tangjia, Zhuhai, Guangdong 519082, P.R. China

Current Position

Name and Address of the Employer: Prof. Dr. Daniel Gruen, Universitaet Sternwarte der Ludwigs-Maximilians-Universitaet Muenchen, Scheinerstr. 1, 81679, Munich, Germany

Start and End Dates of the Contract: 10/2021 - 09/2024

Professional Status: Post-doctoral Researcher

Employment History

10/2021-Present: Post-doctoral researcher, Universitaet Sternwarte Muenchen, Germany

09/2016 - 09/2021: Ph.D. student, Institute for Particle Physics and Astrophysics, ETH Zürich, Switzerland

09/2015 - 12/2016: INAF Associate, INAF Osservatorio Astronomico di Capodimonte, Italy

09/2012 - 12/2012: Didactic laboratory technician and Library Assistant, Dipartimento di Fisica "Ettore Pancini", Università degli Studi di Napoli "Federico II", Italy

Research Interests

Galaxies: clusters: scaling relations and stellar populations of early-type galaxies in clusters at intermediate redshifts

Galaxies: population: forward-modelling of galaxy population in wide-field photometric and spectroscopic surveys, stellar population synthesis, galaxy luminosity function

Methods: statistical: forward-modeling, Approximate Bayesian computation, 2D parametric galaxy light profile fitting, full-spectral fitting and SED fitting

Publication List

- [1] Ragagnin, A. ; Meneghetti, M. ; Bassini, L. ; Ragone-Figueroa, C. ; Granato, G. L.; Despali, G. ; Giocoli, C. ; Granata, G. ; Moscardini, L. ; Bergamini, P. ; Rasia, E. ; Valentini, M. ; Borgani, S. ; Calura, F. ; Dolag, K. ; Grillo, C. ; Mercurio, A. ; Murante, G. ; Natarajan, P. ; Rosati, P. ; Taffoni, G. ; Tornatore, L. ; **Tortorelli, Luca**, *Galaxies in the central regions of simulated galaxy clusters*, Astronomy & Astrophysics, Volume 665, id.A16, 11 pp., September 2022.
- [2] Meneghetti, M. ; Ragagnin, A. ; Borgani, S. ; Calura, F. ; Despali, G. ; Giocoli, C. ; Granato, G. L. ; Grillo, C. ; Moscardini, L. ; Rasia, E. ; Rosati, P. ; Angora, G. ; Bassini, L. ; Bergamini, P.; Caminha, G. B. ; Granata, G. ; Mercurio, A. ; Metcalf, R. B. ; Natarajan, P. ; Nonino, M. ; Venusta Pignataro, G. ; Ragone-Figueroa, C. ; Vanzella, E. ; Acebron, A. ; Dolag, K. ; Murante, G. ; Taffoni, G. ; Tornatore, L. ; **Tortorelli, Luca**; Valentini, M., *The probability of galaxy-galaxy strong lensing events in hydrodynamical simulations of galaxy clusters*, arXiv:2204.09065, April 2022.
- [3] Marques Oliveira, J. ; Sicardy, B. ; Gomes-JÃºnior, A. R. ; Ortiz, J. L. ; [...] **Tortorelli, Luca**; [...] Downs, B., *Constraints on the structure and seasonal variations of Triton's atmosphere from the 5 October 2017 stellar occultation and previous observations*, Astronomy & Astrophysics, Volume 659, id.A136, 43 pp., March 2022.
- [4] Granata, G. ; Mercurio, A. ; Grillo, C. ; **Tortorelli, Luca**; Bergamini, P. ; Meneghetti, M. ; Rosati, P. ; Caminha, G. B. ; Nonino, M., *Improved strong lensing modelling of galaxy clusters using the Fundamental Plane: Detailed mapping of the baryonic and dark matter mass distribution of Abell S1063*, Astronomy & Astrophysics, Volume 659, id.A24, 15 pp., March 2022.
- [5] Berner, P. ; Refregier, A. ; Sgier, R. ; Kacprzak, T. ; **Tortorelli, Luca**; Monaco, P., *Rapid Simulations of Halo and Subhalo Clustering*, arXiv:2112.08389, December 2021.
- [6] **Tortorelli, Luca**; Siudek, M.; Moser, B.; Kacprzak, T.; Berner, P.; Refregier, A.; Amara, A.; Garc  a-Bellido, J.; Cabayol, L.; Carretero, J.; Castander, F. J.; De Vicente, J.; Eriksen, M.; Fernandez, E.; Gaztanaga, E.; Hildebrandt, H.; Joachimi, B.; Miquel, R.; Sevilla-Noarbe, I.; Padilla, C.; Renard, P.; Sanchez, E.; Serrano, S.; Tallada-Cresp  , P.; Wright, Angus H., *The PAU Survey: Measurement of Narrow-band galaxy properties with Approximate Bayesian Computation*, Journal of Cosmology and Astroparticle Physics, Volume 2021, Issue 12, id.013, 45 pp., December 2021.
- [7] Mercurio, A. ; Rosati, P. ; Biviano, A. ; Annunziatella, M. ; Girardi, M. ; Sartoris, B. ; Nonino, M. ; Brescia, M. ; Riccio, G. ; Grillo, C. ; Balestra, I. ; Caminha, G. B. ; De Lucia, G. ; Gobat, R. ; Seitz, S. ; Tozzi, P. ; Scodeggio, M. ; Vanzella, E. ; Angora, G. ; Bergamini, P. ; Borgani, S. ; Demarco, R. ; Meneghetti, M. ; Strazzullo, V. ; **Tortorelli, Luca**; Umetsu, K. ; Fritz, A. ; Gruen, D. ; Kelson, D. ; Lombardi, M. ; Maier, C. ; Postman, M. ; Rodighiero, G. ; Ziegler, B., *CLASH-VLT: Abell S1063. Cluster assembly history and spectroscopic catalogue*, Astronomy & Astrophysics, Volume 656, id.A147, 24 pp., December 2021.
- [8] Johnston, H.; Joachimi, B.; Norberg, P.; Hoekstra, H.; Eriksen, M.; Fortuna, M. C.; Manzoni, G.; Serrano, S.; Siudek, M.; **Tortorelli, Luca**; Cabayol, L.; Carretero, J.; Casas, R.; Castander, F.; Fernandez, E.; Garc  a-Bellido, J.; Gaztanaga, E.; Hildebrandt, H.; Miquel, R.; Padilla, C.; Sanchez, E.; Sevilla-Noarbe, I.; Tallada-Cresp  , P., *The PAU Survey: Intrinsic alignments and clustering of narrow-band photometric galaxies*, Astronomy & Astrophysics, Volume 646, id.A147, 23 pp., February 2021.
- [9] **Tortorelli, Luca**; Fagioli, M.; Herbel, J.; Amara, A.; Kacprzak, T.; Refregier, A., *Measurement of the B-band galaxy Luminosity Function with Approximate Bayesian Computation*, Journal of Cosmology and Astroparticle Physics, Issue 09, article id. 048, September 2020.
- [10] Fagioli, M.; **Tortorelli, Luca**; Herbel, J.; Z  rcher, D.; Refregier, A.; Amara, A., *Spectro-imaging forward model of red and blue galaxies*, Journal of Cosmology and Astroparticle Physics, Issue 06, article id. 050, June 2020.
- [11] Cabayol-Garcia, L.; Eriksen, M.; Alarc  n, Alex; Amara, A.; Carretero, J.; Casas, R.; Castander, F. J.; Fern  ndez, E.; Garc  a-Bellido, J.; Gaztanaga, E.; Hoekstra, H.; Miquel, R.; Neissner, C.; Padilla, C.; S  nchez, E.; Serrano, S.; Sevilla-Noarbe, I.; Siudek, M.; Tallada, P.; **Tortorelli, Luca**, *The PAU Survey: background light estimation with deep learning techniques*, Monthly Notices of the Royal Astronomical Society, Volume 491, Issue 4, p.5392-5405, February 2020.
- [12] Eriksen, M.; Alarcon, A.; Gaztanaga, E.; Amara, A.; Cabayol, L.; Carretero, J.; Castander, F. J.; Crocce, M.; Delfino, M.; De Vicente, J.; Fernandez, E.; Fosalba, P.; Garcia-Bellido, J.; Hildebrandt, H.; Hoekstra, H.; Joachimi, B.; Norberg, P.; Miquel, R.; Padilla, C.; Refregier, A.; Sanchez, E.; Serrano, S.; Sevilla-Noarbe, I.; Tallada, P.; Tonello, N.;

Tortorelli, Luca, *The PAU Survey: early demonstration of photometric redshift performance in the COSMOS field*, Monthly Notices of the Royal Astronomical Society, Volume 484, Issue 3, p.4200-4215, April 2019.

[13] Tonello, N.; Tallada, P.; Serrano, S.; Carretero, J.; Eriksen, M.; Folger, M.; Neissner, C.; Sevilla-Noarbe, I.; Castander, F. J.; Delfino, M.; De Vicente, J.; Fernandez, E.; Garcia-Bellido, J.; Gaztanaga, E.; Padilla, C.; Sanchez, E.; **Tortorelli, Luca**, *The PAU Survey: Operation and orchestration of multi-band survey data*, Astronomy and Computing, Volume 27, article id. 171, April 2019.

[14] Cabayol, L.; Sevilla-Noarbe, I.; Fernández, E.; Carretero, J.; Eriksen, M.; Serrano, S.; Alarcón, A.; Amara, A.; Casas, R.; Castander, F. J.; de Vicente, J.; Folger, M.; García-Bellido, J.; Gaztanaga, E.; Hoekstra, H.; Miquel, R.; Padilla, C.; Sánchez, E.; Stothert, L.; Tallada, P.; **Tortorelli, Luca**, *The PAU survey: star-galaxy classification with multi narrow-band data*, Monthly Notices of the Royal Astronomical Society, Volume 483, Issue 1, p.529-539, February 2019.

[15] Stothert, L.; Norberg, P.; Baugh, C. M.; Alarcon, A.; Amara, A.; Carretero, J.; Castander, F. J.; Eriksen, M.; Fernandez, E.; Fosalba, P.; Garcia-Bellido, J.; Gaztanaga, E.; Hoekstra, H.; Padilla, C.; Refregier, A.; Sanchez, E.; **Tortorelli, Luca**, *The PAU Survey: spectral features and galaxy clustering using simulated narrow-band photometry*, Monthly Notices of the Royal Astronomical Society, Volume 481, Issue 3, p.4221-4235, December 2018.

[16] **Tortorelli, Luca**; Della Bruna, L.; Herbel, J.; Amara, A.; Refregier, A.; Alarcon, A.; Carretero, J.; Castander, F. J.; De Vicente, J.; Eriksen, M.; Fernandez, E.; Folger, M.; García-Bellido, J.; Gaztanaga, E.; Miquel, R.; Padilla, C.; Sanchez, E.; Serrano, S.; Stothert, L.; Tallada, P. Tonello, N., *The PAU Survey: a forward modeling approach for narrow-band imaging*, Journal of Cosmology and Astroparticle Physics, Issue 11, article id. 035, November 2018.

[17] Fagioli, M.; Riebartsch, J.; Nicola, A.; Herbel, J.; Amara, A.; Refregier, A.; Chang, C.; Gamper, L.; **Tortorelli, Luca**, *Forward modeling of spectroscopic galaxy surveys: application to SDSS*, Journal of Cosmology and Astroparticle Physics, Issue 11, article id. 015, November 2018.

[18] **Tortorelli, Luca**; Mercurio, A.; Paolillo, M.; Rosati, P.; Gargiulo, A.; Gobat, R.; Balestra, I.; Caminha, G. B.; Annunziatella, M.; Grillo, C.; Lombardi, M.; Nonino, M.; Rettura, A.; Sartoris, B.; Strazzullo, V., *The Kormendy relation of galaxies in the Frontier Fields clusters: Abell S1063 and MACS J1149.5+2223*, Monthly Notices of the Royal Astronomical Society, Volume 477, Issue 1, p.648-668, June 2018.

Relevant Experience

Observing Experience

Observing nights: 05/2019, William Herschel Telescope, PAUCam, 4n

Observing nights: 03/2018, William Herschel Telescope, PAUCam, 3n

Observing proposal: ESO 0102.A-0640, UT4-Yepun, MUSE, 12h, Co-I

Observing proposal: ESO 0101.A-0470, VST, OMEGACAM, 48h, Co-I

Observing proposal: ESO 099.B-0242, UT4-Yepun, MUSE, 1n, Co-I

Data reduction: basic experience with the ESO data reduction pipeline for the MUSE instrument

Collaborations

4MOST Science collaboration, 08/2022 - Present

Dark Energy Science collaboration (DESC-LSST), 10/2021 - Present

PAU Survey collaboration, P.I. E. Gaztanaga, 03/2017 - Present

CLASH-VLT collaboration, P.I. Piero Rosati, 09/2015 - Present

Talks and Seminars

Updates talk: "DESC Sprint week", hybrid format, University of Michigan, Ann Arbor, USA, 10/2022

Updates talk: "DESC Chicago Collaboration meeting", Chicago, USA, 08/2022

Contributed talk: "Zooming into Dark Matter and Proto-galaxies with gravitational lensing", Sexten, Italy, 07/2022

Contributed talk: "Likelihood-free inference (LFI) in Paris", Paris, France, 04/2022

Contributed talk: "CLASH-VLT collaboration meeting", hybrid format, Florence, Italy, 10/2021

Poster talk: "The 9th KIAS Workshop on Cosmology and Structure Formation" conference, online meeting, 11/2020

Pre-recorded talk: "Galaxy Formation & Evolution in the Era of the Nancy Grace Roman Space Telescope" conference, online meeting, 10/2020

Pre-recorded talk: "Cosmology from Home" conference, online meeting, 08/2020

Contributed talk: "PAUS workshop", Institute of Space Sciences (IEEC-CSIC), Barcelona (Spain), 02/2020

Invited talk: Seminar, Stanford University, Stanford (USA), 10/2019

Contributed talk: "Tracing Cosmic Evolution with Clusters of Galaxies", Sexten Center for Astrophysics, Sexten (Italy), 07/2019

Contributed talk: "Swiss Cosmology Days", University of Zurich, Zurich (Switzerland), 07/2019

Contributed talk: "Swiss-South Africa meeting", ETH Zurich, Zurich (Switzerland), 06/2019

Contributed talk: "Artificial Intelligence methods in Cosmology", Centro congressi Stefano Franscini, Ascona (Switzerland), 06/2019

Contributed talk: "Cosmology on Safari", University of KwaZulu-Natal, Hluhluwe (South Africa), 03/2019

Contributed talk: "CLUSTER2 meeting", Complesso dei SS. Marcellino e Festo, Università degli Studi di Napoli "Federico II", Naples (Italy), 09/2018

Contributed talk: "PAUS workshop", University College London, London (UK), 05/2018

Contributed talk: "PAUS workshop", Institute of Space Sciences (IEEC-CSIC), Barcelona (Spain), 10/2017

Contributed talk: "MUSE Busy week", Rathenow (Germany), 11/2016

Scientific Community Work

Master student supervisor: Benjamin Csizi, Universitaet Sternwarte Muenchen, Munich, Germany, 10/2021 - 08/2022

Bachelor student supervisor: Paulina Contreras, Universitaet Sternwarte Muenchen, Munich, Germany, 04/2021 - 08/2022

Semester student supervisor: Michael Blanc, ETH Zürich, Zürich, Switzerland, 09/2020 - 12/2020

Semester student supervisor: Pascal Hitz, ETH Zürich, Zürich, Switzerland, 03/2020 - 06/2020

Telescope Operator: ETH Zürich 50cm Telescope for teaching, outreach and research, managing operations and designing new experiments for students, 09/2017 - 08/2020

Master student supervisor: Peter Strattman, ETH Zürich, Zürich, Switzerland, 03/2019 - 09/2019

Semester student supervisor: Beatrice Moser (currently PhD student in Prof. Refregier's group), ETH Zürich, Zürich, Switzerland, 03/2018 - 06/2018

Outreach activity: "ETH Scientifica", ETH Zürich, Zürich, Switzerland, 09/2017

Outreach activity: "Città della Scienza", Naples, Italy, 05/2015

Outreach activity: "Astrokids", INAF Osservatorio Astronomico di Capodimonte, Naples, Italy, 04/2015

Outreach activity: "MSC Lirica Cruise", Italy, 10/2014

Outreach activity: "Città della Scienza", Naples, Italy, 05/2014

Outreach activity: "European Researcher's Night", INAF Osservatorio Astronomico di Capodimonte, Naples, Italy, 09/2013

Outreach activity: "Città della Scienza", Naples, Italy, 05/2013

Additional Research Experience

Course/Workshop: "Deep Learning meets (Astro)physics", ETH Zürich, Zürich, Switzerland, 01/2020

Science Visit: KIPAC, Stanford University, Stanford, USA, scope of the visit was interacting and building new collaborations with the group of Risa Wechsler during Prof. Refregier semester visit in Stanford, 10/2019

Course/Workshop: "Machine Learning for High Energy Physics (ML4HEP)", University of Zürich, Zürich, Switzerland, 02/2019

Summer school: "Russian-Italian School on Astrophysics and Cosmology", Dubna University, Dubna, Russia, 09/2015

Summer school: "Astrophysical Probes of Fundamental Physics", University of Ferrara, Ferrara, Italy, 09/2015

Computing Experience

Coding experience: ability to design and run multi-cores parallel jobs on high performance computing facilities

Coding experience: advanced experience of [Python](#) programming language, used for every day work and data analysis of astronomical data

Coding experience: basic experience of [C](#) and [C++](#) programming language acquired during university courses

Coding experience: basic experience of Structured Query Language ([SQL](#))

Programming experience: advanced experience in using [GALFIT](#), [GALFITM](#), [Source Extractor](#), [SCAMP](#), [SWarp](#), [DrizzlePac](#), [P](#) packages, [UFig](#), [pPXF](#), [Python-FSPS](#), [Prospector](#), [CIGALE](#)

Coding experience: good experience of [Github](#)/[Gitlab](#) repo control acquired during everyday work

Coding experience: good experience of [Tensorflow](#) and [Keras](#), used for astronomical data modelling

Additional Information

Courses/lectures given: Teaching assistant for "Astrophysical Lab" class, Universitaet Sternwarte Muenchen, 10/2022 - 01/2023

Courses/lectures given: Teaching assistant for "Observational Methods for Galaxy Surveys" class, Universitaet Sternwarte Muenchen, 04/2022 - 07/2022

Courses/lectures given: Teaching assistant for "Astrophysical Lab" class, Universitaet Sternwarte Muenchen, 10/2021

- 01/2022

Courses/lectures given: Teaching assistant for "Cosmological Probes" class, ETH Zürich, Zürich, Switzerland 09/2020 - 12/2020

Courses/lectures given: Telescope operator and Teaching assistant for "Advanced Physics Laboratory" class, ETH Zürich, Zürich, Switzerland 09/2017 - 08/2020

Courses/lectures given: Teaching assistant for "Beginners Physics Laboratory" class, ETH Zürich, Zürich, Switzerland 09/2016 - 08/2017

Press release: about work as Telescope operator for "Advanced Physics Laboratory" class, <https://www.phys.ethz.ch/news-and-events/d-phys-news/2019/09/studierende-beobachten-extrasolare-planetten.html>, ETH Zürich, Zürich, Switzerland, 09/2019

Languages

Italian: Mother tongue

English: Very fluent, used for everyday research work, available certificates: Trinity College GESE 9

French: basic understanding of written texts, available certificates: A2 Level (DELF A2)

German: basic level, available certificates: A1 Level

Other skills

During my activity as a Science Communicator, I had the chance to improve my ability to communicate and explain science concepts to adults and children, both in English and Italian. I also improved my ability to organize scientific events in a few time and with few money, using poor materials and lots of creativity.

I was responsible and operator for three years of the ETH 50-cm telescope in Zürich (<https://ipa.phys.ethz.ch/education/advanced-student-laboratory/vp-asl-astro.html>). This telescope is used for teaching, outreach and scientific research. Indeed, ETH students, under mine and another PhD colleague supervision, measured transits of extra-solar hot Jupiters. The results were very encouraging and a press release from ETH was issued about this work (<https://www.phys.ethz.ch/news-and-events/extrasolare-planetten.html>). During the three years as telescope operator, I designed new astronomical experiments for students and I wrote a complete and thorough manual https://ethz.ch/content/dam/ethz/special-interest/phys/particle-physics/quanz-group-dam/documents-old-s-and-p/Courses/vp-asl-astro/hpp_manual_LT.pdf. This manual contains information on how to use the telescope, a data-analysis pipeline I designed with Python and the description of the different experiments.